

COMPUTATION OF AREAS:-

1) What is the carpet area?

- Carpet area is the area enclosed within the walls, actual area to lay the carpet. This area does not include the thickness of walls.
- It is the actual usable area of the apartment.

2) What is the built-up area?

- Built area is the Carpet area plus the thickness of outer walls and the balcony.

3) What is the super built up area?

- Super built up area is the built up area plus proportionate area of common areas such as the lobby, lifts, stairs etc..
- The built up area along with a share of all common areas proportionately divided amongst all unit owners makes up the super built up area.

4) What is meant by BHK?

BHK → Bedroom, Hall, Kitchen...

5) What is meant by 1 acre?

1 Acre = 43,560 sq.ft ; 1 ground = 2400 sq.ft

6) What is Floor Area Ratio?

- The floor area ratio (FAR) is the principal bulk regulation controlling the size of the buildings.

$$\text{FAR} = \frac{\text{Gross floor area}}{\text{Plot size}}$$

$$\frac{\text{Carpet area}}{\text{Plot size}}$$

- * FSI → applicable to balconies
- not applicable to terrace.

7) What are the misconceptions of FAR?

- * If FAR is low, we cannot go for high-rise development.
- * Land value is high & sky-rocketing. Hence FAR has to be increased considerably.

8) Carpet area calculation:-

$$\begin{aligned}\text{Saleable area} &= 1200 \text{ sq.ft} \\ \text{Loading factor} &= 25\% \text{ (or } 1.25\text{)}\end{aligned}$$

Calculation-I:-

$$\begin{aligned}\text{Carpet area} &= 1200 \times (100 - 25)\% \\ &= 1200 \times 75\% \\ &= 900 \text{ sq.ft}\end{aligned}$$

$$\begin{aligned}\text{If you apply } 1.25 \text{ or } 25\% \text{ loading,} \\ \text{saleable area should be } 900 \text{ sq.ft} + \\ 900 \times 0.25 \\ = 1125 \text{ sq.ft.}\end{aligned}$$

Conclusion:- Builder is charging you
for 75 sq.ft more

Calculation-II:-

Find carpet area by measuring dimensions.

$$\begin{aligned}\text{Saleable area} &= 900 \text{ sq.ft} + 900 \times 25\% \\ &= 1125 \text{ sq.ft.}\end{aligned}$$

CORRECT METHOD

9) What is the loading factor?

* Loading factor is a number used for purpose of arriving at
Saleable Area

- * Lift/Elevator area, stair cases, club-house, gymnasium, amenities area.
- * Balcony, terrace areas may also be included in the loading factor.
- * Also parking spaces.

10) What are the factors influencing the cost per sq.ft?

- a) Locality b) shape of the plot c) Type of construction
d) quality of construction e) carpet area f) Facilities

11) What is meant by Setback in a building?

- Open space around the building.
- For building above 15m, ⇒ classified as high rise building
- Delhi Development Authority increased the cap to 17.5m for residential buildings.
- But DFS insists a setback of 6m;
- Stilt floor should be made mandatory.

SET BACK:- — Light and ventilation... of all occupancies

Front Set back — Widening of road

Rear Set back

Side set back

FAR:- — completely scientific... [Master plans of urban areas]

Fire safety, water supply, drainage, sewerage, road width, electric supply.

Stilt floor ≥ 3 m

Typical floor ≤ 2.9 m.

Min size of plot for high rise building = 2000 sq.m. [check] ???

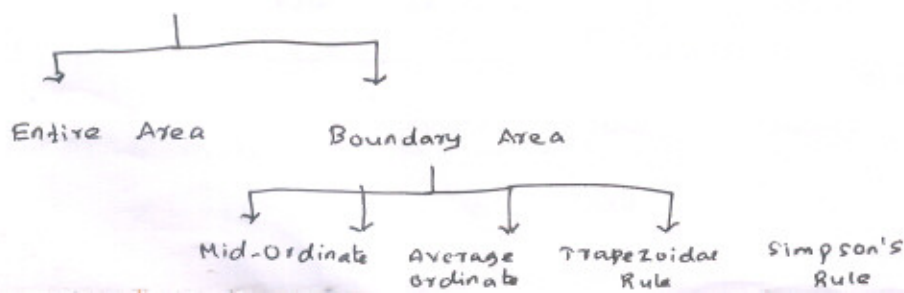
DETERMINATION OF AREAS:

a) From field notes — Cross-staff survey.

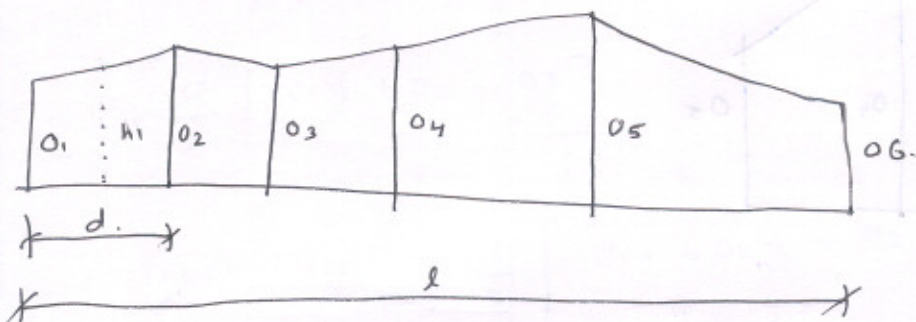
Graphical method

b) From plotted plan or map

Instrumental method...



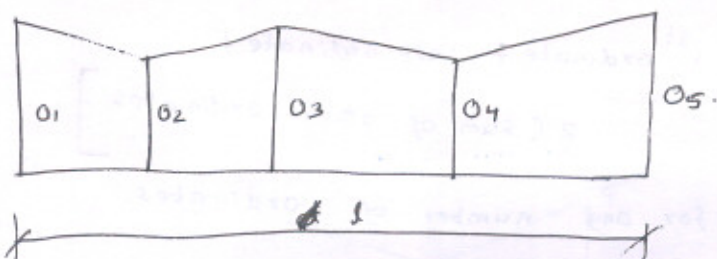
Mid-ordinate Rule:



$$\text{Area of Plot} = (h_1 \times d) + (h_2 \times d) + \dots + (h_n \times d)$$

$$\text{Area} = (\text{Common distance}) \times \text{Sum of the mid-ordinates.}$$

Average Ordinate:-



$n \rightarrow$ no. of divisions;

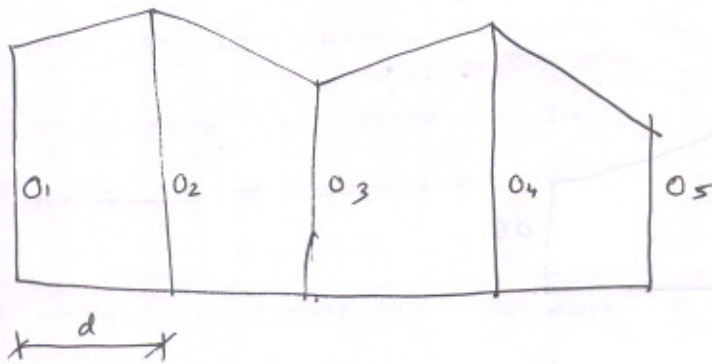
$(n+1) \rightarrow$ no. of ordinates;

$$\text{Area} = \left[\frac{O_1 + O_2 + O_3 + \dots + O_n}{n+1} \right] \times l$$

$$\text{Area} = \left[\frac{\text{Sum of the ordinates}}{\text{no. of ordinates}} \right] \times \text{length of base line}$$

Trapezoidal Rule:

- While applying the trapezoidal rule, boundaries between the ends of ordinates are assumed to be straight. Thus the areas enclosed between the base line and the irregular boundary line are considered as trapezoids...



First area = $\left(\frac{O_1 + O_2}{2}\right) \times d$; Second area = $\frac{O_2 + O_3 \times d}{2}$

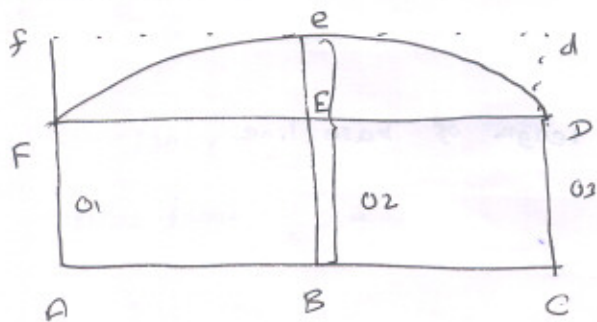
$$\text{Total area} = \frac{d}{2} \left[O_1 + 2O_2 + \dots + 2O_{n-1} + O_n \right]$$

$$= \frac{\text{common distance}}{2} \left[\begin{array}{l} \text{1st ordinate} + \text{last ordinate} + \\ 2(\text{sum of other ordinates}) \end{array} \right]$$

- No limitation; Can be applied for any number of ordinates.

Simpson's Rule:-

- Boundaries between the ends of ordinates are assumed to form an arc of a parabola. Parabolic rule.



Area AFEDC =

Area of trapezium
AFDC +
Area of segment
FEDEF;

Area of trapezium = $\frac{O_1 + O_3}{2} \times 2d$

Area of segment = $\frac{2}{3}$ area of parallelogram FFD

$$= \frac{2}{3} \times Ee \times 2d = \frac{2}{3} \left[O_2 - \frac{O_1 + O_3}{2} \right] \times 2d$$

$$\Delta_1 = \frac{O_1 + O_3}{2} \times 2d + \frac{2}{3} \left[O_2 - \frac{O_1 + O_3}{2} \right] \times 2d$$

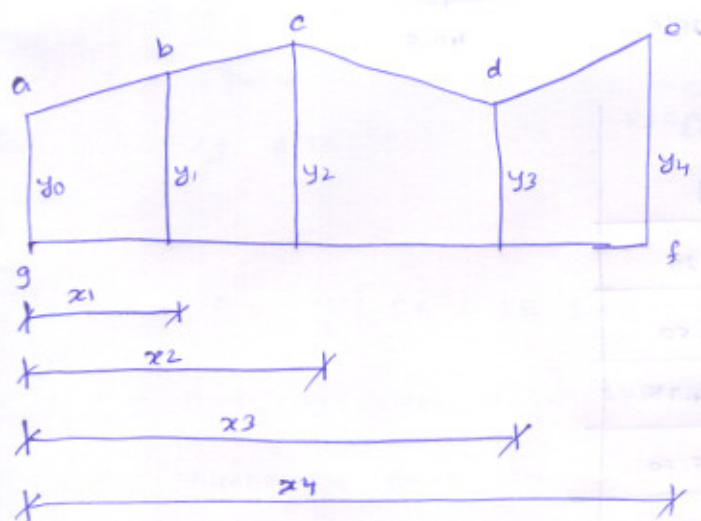
$$= \frac{d}{3} [O_1 + 4O_2 + O_3]$$

$$\therefore \text{Total area} = \frac{d}{3} [O_1 + 4O_2 + 2O_3 + 4O_4 + \dots + O_n]$$

$$= \frac{\text{Common}}{3} [(F.V + L.V) + 4(\text{Even}) + 2(\text{Odd})];$$

Limitation: No. of ordinates is odd.

CO-ORDINATE METHOD OF FINDING AREA:-



Points	Co-ordinates	
	x	y
a	0	y ₀
b	x ₁	y ₁
c	x ₂	y ₂
d	x ₃	y ₃
e	x ₄	y ₄
f	x ₄	0
g	0	0
a	0	y ₀

The co-ordinates are arranged in determinant form as follows.

$$\begin{vmatrix} y_0 & x_1 & y_2 & x_3 & y_4 & x_4 & 0 & 0 & y_0 \\ 0 & x_1 & x_2 & x_3 & x_4 & x_4 & 0 & 0 & 0 \end{vmatrix}$$

Sum of the products along the solid line,

$$\Sigma P = (y_0 x_1 + y_1 x_2 + y_2 x_3 + \dots)$$

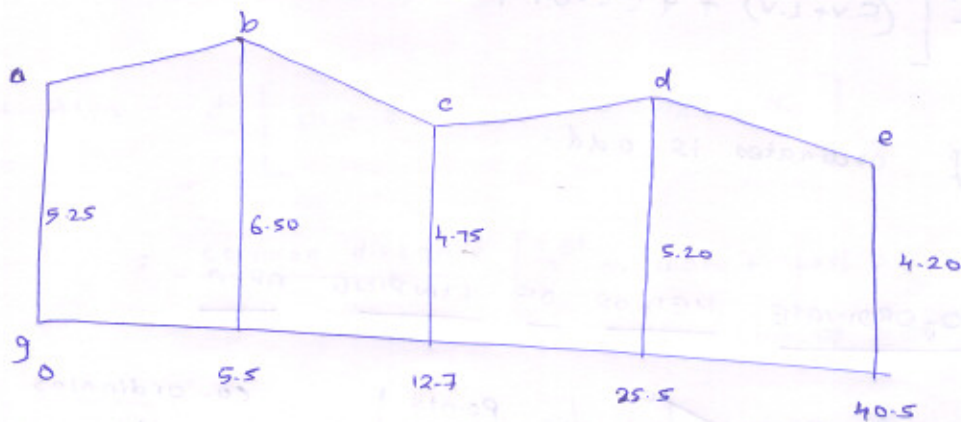
$$\Sigma Q = (0 y_1 + x_1 y_2 + \dots + 0 y_0) \quad [\text{sum of products along the dotted line}]$$

$$\text{Required Area} = \frac{1}{2} [\Sigma p - \Sigma q] \left[\frac{1}{2} + b.s \times \frac{20+50}{5} = 10 \right]$$

PROBLEMS :-

- 1) The following perpendicular offsets were taken from a chain line to a hedge.

Chainage (m)	0	5.5	12.7	25.5	40.5
Offset (m)	5.25	6.50	4.75	5.20	4.20



Point	Coordinates	
	x	y
a	0	5.25
b	5.5	6.50
c	12.7	4.75
d	25.5	5.20
e	40.5	4.20
f	40.5	0
g	0	0
q	0	5.25

$\frac{5.25}{0}$	$\frac{6.50}{5.50}$	$\frac{4.75}{12.7}$	$\frac{5.20}{25.5}$	$\frac{4.20}{40.5}$	$\frac{0}{40.5}$	$\frac{0}{0}$	$\frac{5.25}{0}$
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$$\Sigma p = 613.26$$

$$\Sigma q = 199.27$$

$$\text{Required Area} = 206.995 \text{ m}^2$$

INSTRUMENTAL METHOD:

- Area obtained from Planimeter is more accurate than that obtained by the graphical method.
- Amsler Polar planimeter is the most commonly used planimeter.

Parts of planimeter:-

- Tracing arm
- Anchor arm
- Adjustable support
- Fulcrum point
- Measuring wheel
- Index mark
- Vernier scale
- Pivot point
- Counting disc

Wheel $\rightarrow$ Divided into 100 divisions

Vernier $\rightarrow$ 10 divisions

Wheel is geared to counting disc

[10 revolutions of wheel; one division in disc]

Wheel $\rightarrow$ tenth & hundredths

Vernier $\rightarrow$ thousandths

planimeter can show the reading upto 4 digits } 1.125;
1.124;

$$\text{Area } A = M [FR - IR \pm 10N + C]$$

M - Multiplier given in the table;

N - Number of times the zero mark of the dial passes the index mark

C - Constant given in the table

FR - Final reading

IR - Initial reading

$N > 0$; If zero of the dial passes the index mark in a clock wise direction

$N < 0$ If zero of the dial passes the index mark in a counter clock wise direction.

C is added only when the anchor point is inside the figure.

ZERO CIRCLE OF THE PLANIMETER:

- When the tracing point is moved along the circle without rotation of the wheel (when the wheel slides without any change in reading) the circle is known as the "zero circle" or "circle of correction".

$$\text{Area of zero circle } A = M \times C$$

$$\text{Area of zero circle} = \text{Actual area} - \text{Area computed by Planimeter}$$

NOTE: The anchor point should preferably be kept outside the figure to avoid "additive constant".

Area of zero circle,

$$A = \pi R_1^2 = \pi (L^2 - 2LL_1 + R^2)$$

where $R_1 = \sqrt{L^2 + 2LL_1 + R^2}$

4. From multiplier and constant:

Area of zero circle, $A = M \times C$

where M = multiplier value given in table

C = constant given in table

5. By planimeter:

A geometrical figure is considered whose actual area is known. After this, the area of the figure is computed by the planimeter.

Then,

$$\text{Area of zero circle} = \text{Actual area} - \text{Area computed by planimeter}$$

Point to be Remembered while Using the Planimeter

1. The map must be placed over a horizontal plane.
2. The anchor point should preferably be kept outside the figure to avoid additive constant.
3. The area of the figure should be measured twice from different starting points.
4. If the area is large, it should be divided into a number of sections; the area of each section may be calculated separately by taking outside poles, and then added to obtain the total area.
5. The initial reading may be set to zero for the sake of simplicity.
6. The tracing point should be moved gently and exactly along the boundary line.
7. The map should not be folded.
8. The surface of the map should be smooth.

7.8 WORKED-OUT PROBLEMS

Problem 1 The following readings were recorded by a planimeter with the anchor point inside the figure:

$$IR = 9.377, FR = 3.336, M = 100 \text{ cm}^2, \text{ and } C = 23.521$$

Calculate the area of the figure when it is observed that the zero mark of the dial passed the index mark once in the anticlockwise direction.

Solution Given data:

$$IR = 9.377$$

$$M = 100 \text{ cm}^2$$

$$FR = 3.336$$

$$C = 23.521$$

From the expression, $A = M(FR - IR \pm 10N + C)$

we get, $A = 100(3.336 - 9.377 - 10 \times 1 + 23.521) = 748 \text{ cm}^2$

Problem 2 The following particulars were noted while measuring the area of a figure with a planimeter.

- (a) IR and FR were 8.652 and 6.798 respectively.
- (b) The tracing arm was set to the natural scale.
- (c) The zero of the dial passed the index mark once in the anticlockwise direction.
- (d) Constant $C = 20$.
- (e) Scale of the map is $1 \text{ cm} = 10 \text{ m}$.
- (f) The anchor point was inside the figure.

Calculate the area of the figure.

Solution Given data

$$IR = 8.652$$

$$FR = 6.798$$

'Natural scale' means that $M = 100 \text{ cm}^2$

$$C = 20$$

$$N = -1 \quad (\text{for anticlockwise rotation})$$

$$\text{Scale: } 1 \text{ cm} = 10 \text{ m}$$

Area of the figure

$$A = M(FR - IR - 10 \times N + C)$$

$$= 100(6.798 - 8.652 - 10 \times 1 + 20) = 814.6 \text{ cm}^2$$

Since the scale is $1 \text{ cm} = 10 \text{ m}$,

$$1 \text{ cm}^2 = 100 \text{ m}^2$$

$$\text{Required area} = 814.6 \times 100 = 81460 \text{ m}^2$$

Problem 3 The area of an irregular figure was measured with a planimeter having the anchor point outside the figure. The initial and final readings were 4.855 and 8.754 respectively. The tracing arm was set to the natural scale. The scale of the map was $1 \text{ cm} = 5 \text{ m}$. Find the area of the figure.

Solution Given data:

$$IR = 4.855$$

$$FR = 8.754$$

$$M = 100 \text{ cm}^2 \text{ (natural scale)}$$

$N = 0$ (because there is no comment about the crossing of the index mark)

$$\text{Scale: } 1 \text{ cm} = 5 \text{ m}$$

$$C = 0 \text{ (anchor point outside)}$$

So,

$$\begin{aligned}\text{Area } A &= M(FR - IR) \\ &= 100(8.754 - 4.855) \\ &= 389.9 \text{ cm}^2 \\ \text{Required area} &= 389.9 \times 25 = 9,747.5 \text{ m}^2 \quad (\because 1 \text{ cm}^2 = 25 \text{ m}^2)\end{aligned}$$

Problem 4 A field of 0.16 hectare area was plotted to a scale of 1 cm = 4 m. The area of this plan was measured with a certain setting of a planimeter, and the following details were obtained, when the anchor point was outside the area:

$$IR = 3.415, FR = 4.415, N = 0$$

Another area was traced with the same setting of the planimeter by keeping the anchor point inside the area, and the following observations were obtained:

$$FR - IR = 2.250 \text{ and } N = -1$$

The value of the additive constant for this scale was noted as 21.22. Find the area in the second case, and the area of the zero circle.

Solution First Case

Given data:

$$\begin{aligned}IR &= 3.415, FR = 4.115, N = 0, M = ? \\ C &= 0 \quad (\text{anchor point outside})\end{aligned}$$

$$\text{Scale: } 1 \text{ cm} = 4 \text{ m}$$

$$1 \text{ cm}^2 = 16 \text{ m}^2$$

$\therefore$

$$\begin{aligned}\text{Area on the ground} &= 0.16 \times 10,000 \text{ (1 hectare} = 10,000 \text{ m}^2) \\ &= 1,600 \text{ m}^2\end{aligned}$$

$$\text{Area of map} = \frac{1,600}{16} = 100 \text{ cm}^2$$

So,

$$A = 100 \text{ cm}^2 \quad (\text{measured area on plan})$$

From the relation,

$$\begin{aligned}A &= M(FR - IR) \\ 100 &= M(4.415 - 3.415) = M \times 1 \\ M &= 100\end{aligned}$$

Second Case

Given data:

$$\begin{aligned}FR - IR &= 2.250 \\ C &= 21.22 \\ N &= -1\end{aligned}$$

$$\begin{aligned}\text{Required area, } A &= M(FR - IR - 10 \times 1 + 21.22) \\ &= 100(2.250 - 10 + 21.22) = 1,347 \text{ cm}^2\end{aligned}$$

$$\text{Area of zero circle} = M \times C$$

Problem 5 The following readings were obtained while describing the perimeter of a rectangle of size 15 cm $\times$ 10 cm with the anchor point inside the rectangle and tracing arm set to the natural scale ($M = 100$): $IR = 0.686$ and $FR = 9.976$. The zero of the counting disc passed the index mark twice in the anti clockwise direction. Find the area of the zero circle.

$$\text{Solution Area of rectangle} = 15 \times 10 = 150 \text{ cm}^2$$

$$IR = 0.686, FR = 9.976, N = -2, M = 100$$

$$C = ?$$

$$\begin{aligned}\text{Measured area of rectangle} &= M(FR - IR - 10N + C) \\ &= 100(9.976 - 0.686 - 10 \times 2 + C) \\ &= 100(-10.710 + C)\end{aligned} \quad (1)$$

$$\text{Again, measured area} = 150 \text{ cm}^2 \quad (2)$$

From (1) and (2),

$$\begin{aligned}100(-10.710 + C) &= 150 \\ C &= 1.5 + 10.710 = 12.21\end{aligned}$$

$$\begin{aligned}\text{Hence, area of zero circle} &= M \times C \\ &= 100 \times 12.21 = 1,221 \text{ cm}^2\end{aligned}$$

Problem 6 Calculate the area of the zero circle with the following data:

IR	FR	Position of anchor point	Remark
7.775	4.825	Outside the figure	The zero crosses the index mark once clockwise
2.325	8.755	Inside the figure	The zero crosses the index mark twice anticlockwise

Solution First Case:

$$\begin{aligned}N &= 1 \\ C &= 0 \quad (\text{as anchor point is outside the figure}) \\ M &= 100\end{aligned}$$

$$\begin{aligned}\text{Area of figure} &= M(FR - IR + 10N) \\ &= 100(4.825 - 7.775 + 10 \times 1) \\ &= 100 \times 7.050 = 705.0 \text{ cm}^2\end{aligned} \quad (1)$$

Second Case:

$$\begin{aligned}M &= 100 \\ N &= -2\end{aligned}$$

$$C = ?$$

$$\text{Area of figure} = M(FR - IR - 10N + C)$$

$$= 100(8.755 - 2.325 - 10 \times 2 + C)$$

Equating (1) and (2),

$$100(-13.570 + C) = 705.0$$

$$C = 7.050 + 13.570 = 20.620$$

Hence, Area of zero circle = $M \times C = 100 \times 20.620$
 $= 2062 \text{ cm}^2$

7.9 DIGITAL PLANIMETER

The digital planimeter is a computerized instrument with pole arms or rollers. It gives directly the reading of an area calculated in sq. inches, sq. metres, sq. kilometres, etc. The storage unit is used for downloading data into a computer. The unique facility is that it provides adding areas, accumulating measurements and average multiple measurements. The digital planimeter requires initial setting for unit and scale. Therefore, it never requires a multiplying constant like an ordinary planimeter for calculation of area.

The digital planimeter can measure area with an accuracy of plus/minus 0.2% of the whole scale.

The PLACOM N series Planimeter is digital. Measurement of area is done here by a 6-digit pulse count. The measuring capability of a digital Planimeter is 100 times larger than an ordinary planimeter.



Photo Digital Planimeter

Short-Answer Questions for Viva

1. State the trapezoidal rule. What are the considerations and limitations of this rule?

"To the sum of the first and the last ordinate, twice the sum of the intermediate ordinates is added. This total sum is multiplied by the common distance. Half of this product is the required area". This is the trapezoidal rule.

The boundaries between the ends of ordinates are assumed to be straight lines. There is no limitation in this rule. It can be applied at any number of ordinates.

2. State Simpson's rule. What are the considerations and limitations of this rule?

To the sum of the first and the last ordinate, four times the sum of even ordinates and twice the sum of odd ordinates are added. This total sum is multiplied by the common distance. One-third of this product is the required area." This is Simpson's rule.

The boundary between the ordinates is assumed to form an arc of a parabola. To apply this rule, the number of ordinates must be odd.

3. What is a planimeter?

It is an instrument for measuring the area of a field from the map.

4. What is a zero circle?

When a circle is described by the tracing point without a change in reading in the measuring wheel, then that circle is known as the zero circle.

5. Give the simplest method for finding the area of a zero circle from the manufacturer's table.

$$\text{Area of zero circle} = M \times C$$

where, M = multiplier

C = constant

The values of both M and C are available in the table.

6. What is the need of finding the area of the zero circle?

When the anchor point is inside the figure, the computed area does not cover the whole area. It is less by the area of the zero circle. In that case, the area of the zero circle is added to the computed area to obtain the actual area.

Exercises

1. (a) State the trapezoidal rule and Simpson's rule. What is the limitation of Simpson's rule.
 (b) The following perpendicular offsets were taken from a chain line to a hedge:

Distance (m)	0	6	12	18	24	30	36
Offset (m)	5.40	4.50	3.60	2.70	1.80	2.25	3.15

Calculate the area enclosed between the chain line and the offsets by (i) trapezoidal rule, and (ii) Simpson's rule.
 (Ans. 114.75 m², 114.3 m²)

2. The following perpendicular offsets were taken from a chain line to a hedge:

Distance (m)	0	5	10	15	20	30	40	50	65	80
Offset (m)	3.40	4.25	2.60	3.70	2.90	1.80	3.20	4.50	3.70	2.80

Calculate the area by (i) trapezoidal rule, and (ii) Simpson's rule

(Ans. 261.75 m², 265.5 m²)

3. The following perpendicular offsets were taken at 10 m intervals from a chain line to an irregular boundary line:

3.10, 4.20, 5.35, 6.45, 7.15, 8.25, 7.95 and 5.20 m

Find the area by (i) trapezoidal rule, and (ii) Simpson's rule.

(Ans. 436.50 m², 438.95 m²)

$$\begin{aligned}
 III &= 9 \left[\frac{3.0 + 3.9}{2} + 3.3 \right] \\
 &= 9 \left[\frac{6.9}{2} + 3.3 \right] \\
 &= 9 [3.45 + 3.3] = 9 \times 6.75 = 60.75 \text{ m}^2
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Total area enclosed} &= I + II + III \\
 &= 57.6 + 97.2 + 60.75 = 215.55 \text{ m}^2 \quad \text{Ans.}
 \end{aligned}$$

Example 8.3. A series of offsets were taken from a chain line to a curved boundary line at an interval of 10 m in the following order :

0, 2.85, 3.95, 6.45, 8.60, 8.90, 5.25, 0, metres. Calculate the area between the chain line and the curved boundary line by the Simpson's rule.

Solution. (Fig. 8.12).

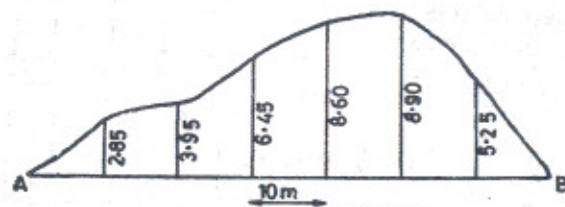


Fig. 8.12.

Note. In order to apply Simpson's rule, the number of offsets should be odd. In this problem, number of offsets is even. Hence, the last offset is ignored for calculating the area of the remaining portion and the area between last two offsets is calculated by trapezoidal rule and added, to get the required area.

$$\begin{aligned}
 \therefore \text{Area} &= \frac{10}{3} [0 + 5.25 + 2 (3.95 + 8.60) + 4 (2.85 + 6.45 + 8.90)] \\
 &\quad + \frac{10}{2} (0 + 5.25) \\
 &= \frac{10}{3} [5.25 + 25.10 + 72.80] + 26.25 = 343.83 + 26.25 \\
 &= 370.08 \text{ sq. m.} \quad \text{Ans.}
 \end{aligned}$$

Example 8.4. (a) What is Simpson's Rule in the computation of areas of figures? Derive an expression for it.

(a) The following offsets were taken from a chain line to a hedge.

Distance in metres	0	30	60	90	120	150	180
Offsets in metres	9.4	10.8	12.5	10.5	14.5	13.0	7.5

Compute the area included between the chain line, the hedge and end offsets by the Simpson's rule.

(A.M.I.E., May 1975)

Solution.

(a) For statement and derivation of the Simpson's rule, refer to article 8.4. Area

$$\begin{aligned}
 A &= \frac{d}{3} [\text{sum of the first and last offsets} + \text{Twice the sum of the remaining odd offsets} + \text{Four times the sum of remaining even offsets}] \\
 &= \frac{30}{3} [19.4 + 7.5 + 2 (12.5 + 10.5) + 4 (10.8 + 10.5 + 13.0)] \\
 &= 10 (16.9 + 54.0 + 137.2) = 2081 \text{ sq. m.} \quad \text{Ans.}
 \end{aligned}$$

Example 8.5. A plot of land ABCDA has four sides. The sides AB and BC are straight and the sides CD and DA are irregular. The above plot was surveyed by chain and tape by the method of chain surveying, fixing four stations at A, B, C and D and was moved and measured in the clockwise direction. The straight distances measured from one station to the other, were as follows :

AB = 150 m ; BC = 165 m ; CD = 155 m ; DA = 162 m ; AC = 230 m ; The offsets measured from chain lines CD and DA to the irregular boundaries are as under :

Distance from C in (m)	Offsets in (m)
0	0.00 left
30	1.50 "
60	2.00 "
90	2.25 "
120	1.75 "
155	0.00 "
Distance from D in (m)	Offsets in (m)
0	0.00 right
30	1.62 "
60	2.45 "
90	2.30 "
120	1.22 "
162	0.00 "

Calculate the area of the plot ABCDA.

(A.M.I.E. 1979, Winter, 1991 Summer)

Solution. (Fig. 8.13).

We know that the area of a triangle whose sides are a , b and c is given by the equation :

$$A = \sqrt{s(s-a)(s-b)(s-c)} \quad \dots(i)$$

where $2s = a + b + c$.

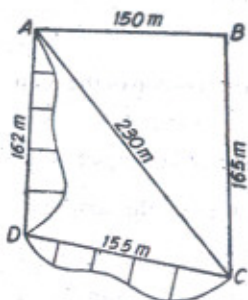


Fig. 8.13.

Area of ΔABC :

$$a = 165 ; b = 230 ; c = 150 ; 2s = 545 ; s = 272.5$$

$$\begin{aligned} \therefore A_1 &= \sqrt{272.5 (272.5 - 165) (272.5 - 230) (272.5 - 150)} \\ &= \sqrt{272.5 \times 107.5 \times 42.5 \times 122.5} \\ &= 12349.49 \text{ sq.m.} \end{aligned}$$

Area of ΔACD

$$a = 155, c = 162, d = 230, 2s = 547, s = 273.5$$

$$\begin{aligned} \therefore A_2 &= \sqrt{273.5 (273.5 - 155) (273.5 - 162) (273.5 - 230)} \\ &= \sqrt{273.5 \times 118.5 \times 111.5 \times 43.5} \\ &= 12537.75 \text{ sq. m.} \end{aligned}$$

Area between irregular boundary and side CD

$$\begin{aligned} A_3 &= \frac{30}{3} \left[0 + 1.75 + 4 (1.50 + 2.25) + 2 \times 2.0 + \frac{0 + 1.75}{2} \times 35 \right] \\ &= 10 [0 - 1.75 + 15.0 + 4.0] + 0.875 \times 35 \\ &= 207.5 + 30.625 \\ &= 238.125 \text{ sq. m.} \end{aligned}$$

Area between irregular boundary and side DA

$$\begin{aligned} A_4 &= \frac{30}{3} [0 + 1.22 + 4 (1.62 + 2.30) + 2 \times 2.45] + \frac{0 + 1.22}{2} \times 42 \\ A_4 &= 10 [0 + 1.22 + 15.68 + 4.9] + 0.61 \times 42 \\ &= 218.0 + 25.62 \\ &= 243.62 \text{ sq. m.} \end{aligned}$$

Area of plot of land ABCDA

$$\begin{aligned} &= \text{Area of } \Delta ABC + \text{Area of } \Delta CDA + A_3 - A_4 \\ &= 12349.49 + 12537.75 + 238.125 - 243.62 \\ &= 24881.745 \text{ sq. m.} \quad \text{Ans.} \end{aligned}$$